PROJECT INTRODUCTION · SEE THE FUTURE 2026 · CIRCULAR ECONOMY

BetterBarter

A verified campus reuse board where students give, sell, lend or swap what they already own — and every confirmed handoff is measured.

betterbarter.vercel.app · github.com/mn3265-commits/betterbarter

Agung Nugroho & Tessa Wong · Columbia University · moh.agungnugroho@gmail.com

THE PROBLEM

A campus throws out most of a year in one week, and buys it back in September.

At move-out, working lamps, fridges, rugs, textbooks and coats go into the same dumpsters — not because students want to discard them, but because listing an item takes longer than carrying it downstairs, and because the alternative is meeting a stranger from a public marketplace off campus. Eight weeks later the next intake buys the identical objects new, often financed on money they do not have.

12.1M t
US furniture and furnishings generated in 2018 — 80.1% landfilled (EPA)

17M t
US textiles generated the same year; 11.3M t landfilled (EPA)

1 week
of May carries most of a residential campus's annual discard

Existing responses are one-week, staffed, one-way donation drives, bound by trucks, storage and volunteer hours. Nothing runs on the ordinary Tuesday when the decision to bin the chair is actually made. **The waste is downstream of a user-experience problem, and that is the part nobody has built for.**

THE SOLUTION

Four ways to keep an object in use, and a code that proves it happened.

Twenty seconds to post

Up to three photos, then a short form: category, condition, size, how you are listing it, and where you want to meet — from a list of real campus places, never a room.

Give, sell, swap — renting next

Ownership is one way to have a thing. A swap keeps two objects in use instead of one; renting is built and announced on the Marketplace, opening once returns are handled properly.

Only your campus sees it

One verified university email per account, enforced in the database. Every school is an isolated board — which removes the stranger problem that stops students listing.

The community is the logistics

A fridge is not a one-person object. Listings can ask for two people or a trolley, and students on the same campus take that work for a fee paid directly between them — no van dispatched, so the transport term stays zero.

Both sides confirm, in person

The exchange happens at a library entrance or a dining hall door — and it is counted only when six digits, split three-and-three across the two phones, are put back together there. Neither app can read the other's half, so the count cannot be raised by one person alone.

WHY IT IS DIFFERENT

Reuse measured at the object, not at the truck.

- **An auditable circularity metric.** Drives report a weight after the fact, in aggregate. BetterBarter records one confirmed event per object, with category and timestamp, agreed by both parties — bottom-up, and defensible line by line.
- **The measurement is the reputation.** The tap that counts the reuse is the tap that raises both people's public handoff count, so the incentive to record impact is intrinsic rather than bolted on for reporting.
- **Heavy things move without a vehicle.** The wall every reuse venture hits is the fridge nobody can carry. A campus already contains the answer: a student with an hour and a trolley, paid directly, with a public carry count of their own.
- **No transport term.** A collection round burns fuel per item — roughly 2.5 kg CO_2e for a 10 km round trip, at the EPA's passenger-car factor, which flatters a van. Two students crossing a campus they were crossing anyway emit nothing extra.
- **An honest model, published in the open.** Measured, estimated and conservative numbers are kept apart; factors take the low end; displacement is held at 0.5; and a borrowed object earns no carbon credit at all, because it comes back.
- **Any university, instantly.** A school with no board gets one the moment its first student signs in — the campus is created, named and marked from a registry of 7,328 academic domains. No waiting list, no launch.

THE IMPACT MODEL · V1.1

Three levels, never blurred.

LEVEL	NUMBER	STATUS
1	Confirmed handoffs — both people present, code joined, one object, one timestamp	**Measured**
2	Mass kept in use = count × typical mass for the category	Estimated
3	Production avoided = mass × low-end cradle-to-gate factor × displacement 0.5	Estimated, conservative

Forecasts use an assumed move-out category mix weighted toward small things rather than an unweighted average, which lowers our own estimate by roughly a third. On that basis a single participating floor — 120 students, three items each — is about **360 objects, 1.2 tonnes kept out of landfill and 2.6 tonnes CO_2e of avoided production**. The full factor table, the mix and the sources are printed on the public site, and the model lives in one file in an open repository.

MARKET

Counted two ways, with the arithmetic shown.

The material in play

TAM	46M objects a year · 159,000 t · 333,000 t CO_2e
SAM	14M objects · 48,000 t
SOM · 3 yr	300,000 objects · 1,000 t

15.4M US undergraduates (NCES) × 3 reusable objects a year, at our published 3.45 kg and 7.2 kg CO_2e per object.

The revenue — campus licences

TAM	$58M a year
SAM	$18M a year
SOM · yr 3	$375k a year

3,896 US degree-granting institutions (NCES 2022–23) × a $15k licence; SAM is the ~1,200 residential four-year campuses; SOM is 25 of them. Placements excluded.

CURRENT STAGE

Working product, pre-pilot.

Deployed and running on a live backend, not a mockup: university sign-in, campus-isolated Postgres with row-level security on every table, realtime chat that refuses to send contact details, photo storage, an hourly freshness check that retires stale listings by itself, a community-rules gate, a pre-post safety check, ratings, reporting with a moderation queue behind it, and rate limits enforced in the database. It is an installable PWA and a full desktop web app. Handoffs are confirmed in person with a six-digit code split across the two phones.

The roll-out runs in four phases: fifty to a hundred Columbia SPS students invited by hand during orientation week, then the School of Professional Studies within a month, the university within six, and other institutions after that — a university email domain already maps to a campus server-side, so a new school's Marketplace exists the moment its first student signs in. The pilot is scoped through the May 2027 move-out, which is the week the thesis is actually tested.

EVIDENCE OF TRACTION

What exists, and what does not.

What exists. A live product at betterbarter.vercel.app, open source at github.com/mn3265-commits/betterbarter. A hardened backend: row-level security on every table, an RLS-hardening pass after the security advisor flagged it, server-side sign-up gating on academic domains, database-level rate limits, a reporting and moderation queue, and a confirmation function that cannot double-count. The impact model published in full with its factor table, its assumed category mix, its displacement rate and every source. All of it built and shipped by a two-person team in five weeks.

What does not. No paying customers. No institutional partnership. No pilot users — two verified Columbia accounts exist and both of them are ours. No conversation has happened yet with Columbia Housing, EcoReps or the Office of Sustainability. We would rather state that plainly than dress it up: the thing we are asking to be judged on is a working product and a defensible method, not a user count we do not have.

HOW IT PAYS FOR ITSELF

Charged before a deal exists, never on the object.

Campus licence

The main line. Housing and sustainability offices already pay to haul this material away and already must report circularity; a licensed campus gets admin tools, the move-out programme and an audited object-level number.

Verified-audience placements

Movers, storage and student services reaching one campus at one fortnight of the year. Paid up front, clearly marked, never mixed into ranking.

Free permanently

Posting, claiming, giving, renting, swapping. No listing fee and no cut of a handoff — a fee on a give-away would be a tax on generosity.

TEAM

Two people, in the open.

Agung Nugroho — Co-Founder, Tech and Product. Designs the Marketplace, writes the code, owns the impact model, runs the pilot. Graduate student and Graduate Assistant at Columbia SPS; professional background in customer-lifecycle and CRM marketing at telecommunications scale, where the daily problem is the same one this product has: getting a very large number of ordinary people to complete one specific action, and measuring honestly whether they did.

Tessa Wong — Co-Founder, Strategy and Community. A sustainable development professional by background. She began her career as a management consultant at Accenture Singapore, before pivoting to sustainable development research and consulting — first for the development sector in Cambodia, then for the private sector in Singapore.

Campus reuse does not fail on technology. It fails on adoption and on trust, which are behaviour and product problems before they are engineering ones.

SCALE

Dense before wide.

Adding a school costs nothing and requires no code: a university email opens a Marketplace, and row-level security keeps every campus isolated. What does not copy is density, so our attention stays on one campus — building by building — until it is useful on an ordinary Tuesday and not only in the last week of May. Distribution then runs through institutions rather than advertising: housing, sustainability offices and student groups already own the move-out problem, and the impact model is the pitch to them.
